Research Data Management in Three Spanish Universities

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Abstract. Research Data Management (RDM) is an important ability required in the global knowledge environment and essential in the university research context. In this work we will present the results of the application of a web survey of faculties and doctoral students from three Spanish universities, with the aim of knowing their current levels of awareness and gaps in different issues of RDM. This study is part of an international survey on "Base Data Literacy" led by Professor Joumana Boustany from Paris Descartes University. The questionnaire was sent to the research academic staff and student research fellows of the three universities. We received a total of 828 responses, 591 of which were completely filled out. In accordance to the results of this survey we have detected a growing need among research academic staff and research students for RDM skills.

Keywords: Research Data Management · Data management plan
Spanish's Universities · Metadata

1 Introduction

Research Data Management (RDM), is defined by Schneider [1] as "a method that enables the integration, curation and interoperability of data created during the scientific process, i.e. the production, access, verification, persistent storage and reuse of this data with the help of adequate and easy-to-use tools in virtual research infrastructures."

In the field of research, within the Horizon 2020 program [2], the European Commission (EC) and the Guidelines for the Regulation on Open Access in Scientific Publications and Research Data [3] define research data as the Information based on actual facts and that can be proved as well as the numerical information that is collected to be analyzed and considered as the basis of a reasoning, a discussion, or a calculation. Statistical results, results of experiments, and observations resulting from field studies, surveys, interviews, and images all constitute examples of research data. Thus, Research Data Management is present in all phases of the research [4] since it includes

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the processes of data creation, planning for reuse, processing, and organization of the same in their structuring and the content analysis, conservation, along with taking into account the security of access, its storage, recovery, public sharing, and reuse. Guidelines on all aspects of the "Participant Portal" were published in 2014 by the European Commission in relation to Horizon 2020 [5].

This reality has led to researchers having to acquire data management capabilities in order to manage huge amounts of data. These capabilities include statistical and computer skills, among others, and the know-how to contextualize the data appropriately in favour of the development of the organization. Leber [6] states that this new task has led to the birth of a new profession that the author calls "data scientist", which is already indispensable in companies as well as in the field of research.

In Spain, today data management in research is a reality that has begun to develop mainly because of the requirements of Horizon 2020, which is beginning to be replicated in national programs.

At the moment, in 2017, Horizon 2020 is a great incentive because all the projects that receive EU funding should develop a data management plan and publish open research data.

As a predecessor of the Spanish situation, we have a report published in 2014 by the Higher Council for Scientific Research (CSIC) [7] on the practices in the management, dissemination, and preservation of data of this institution. In this study, 270 researchers were interviewed and the most relevant conclusions, among others, focused on the insecurity that the interviewees showed regarding control of their own data, the fear of sharing these data, or the idea that the data did not interest others. These results already evidenced the urgency of forming and creating support services for researchers that included the creation, updating, preservation, and dissemination of research data, and for that purpose, university libraries were involved. In this sense, it is important to study Piorun [8], where she argues that these libraries can be very useful since they are experts in the selection of resources, creation of metadata, preservation, organization, and management of data and in providing access to them.

Since 2014, many universities and university libraries have created support services for researchers in relation to data management. One of the most relevant services was created by the Consortium of University Services of Catalonia (CSUC) called "Research Support Working Group", composed of members of all Catalan universities and experts in LIS and technologies of information. This service provides the research groups with a guide for the design of a data management plan (PGD), recommendations for selecting the most appropriate repository for the repository of data, and an open access institutional policy document [9]. Another initiative to support data management, called PaGoDa [10], was developed by the Madroño Consortium of university libraries in the Community of Madrid and the National University of Distance Learning (UNED), that adapted the tool, DMP Online, developed by the Digital Curation Center of the United Kingdom. This consortium also offers support material and advice through each of its member libraries. These initiatives have been very important in Spain, both for the support they provide to researchers and for training in the acquisition of skills in data management.

From the legislative point of view, there are many gaps in our country, one of which refers to data authorship. The law of intellectual property considers authorship in

databases as intellectual creations; however, databases containing raw data do not have intellectual property. Licensure that permits data to be published it is usually done through Creative Commons, recommended by the European Commission in 2017.

At the end of 2016, Professor Joumana Boustany, from Paris Descartes University, invited us to participate in the international survey, "Base Data Literacy", for senior and junior researchers at universities. We felt that the timing was very appropriate because 2017 was to be the key date in promoting the management of research data. The results would help us to appreciate a general framework of the Spanish situation. Therefore, the objective of this study is to collect information about RDM of academic staff and doctoral students from Spanish universities. The analysis of the results obtained will allow us to understand the current levels of awareness and gaps in knowledge that will help us develop appropriate data literacy training for the higher education community.

2 Methodology

We used an on-line survey that was designed by Serap Kurbanoğlu (Hacettepe University, Turkey) and Joumana Boustany (Paris Descartes University, France). The survey was translated to Spanish for better comprehension. The questionnaire contains 24 questions, most of which have closed response options and is divided into two sections. The first gathers demographic information including age, gender and educational and professional experience of the respondents. The second explores the opinions of the respondents on different issues of data management: types of data file used in research, volume of data, how you obtain and use the data, where the data is stored, the Data Management Plan, and training in data management.

In accordance with to the objectives of the study, we selected a sample that included research professors and students research fellows from three Spanish public universities representing the three types of Spanish public universities. First, we included the Complutense University of Madrid (UCM) which was created in the century XVI as an example of a traditional university with a large number of professors (more than 5,000) and students (more than 80,000). The second was the Rey Juan Carlos University of Madrid (URJC) as an example of a new university with a medium capacity in number of students (approximately 30,000) and professors (more than 1,400). And, finally, we included the University of Burgos (UBU) an example of a small university in number of students (more than 6,500) and teachers (about 800). The questionnaire was sent only to the research academic staff and student research fellows of the three universities. The number of questionnaires sent were: UCM: 2,740; URJC: 634; and UBU: 425. Of these, 3,301 were addressed to professors and 498 to student research fellows. We received a total of 828 questionnaires, although only 591 were fully completed. The questionnaires received by each University were: 470 of the UCM (response rate 17.1%); 60 of the URJC (response rate 9.5%); 61 of the UBU (response rate 14.3%).

The research was conducted between the months of February and March 2016. Closed-ended questions were analyzed using descriptive statistics including frequency counts and percentages. The data was analyzed statistically with IBM SPSS Statistics.

3 Results

3.1 Respondents Profile

The majority of respondents were research academic staff (84.6%, n = 501) and 15.2% (n = 90) were research students.

As shown in Table 1 the majority of respondents (82.5%) were men older than 36 years of age.

Table 1. Demographic data of the respondents

	Number	Percentage
Gender		
Female	269	45.5
Male	316	53.5
Age		
18–25	17	2.9
26–35	85	14.4
36–45	159	26.9
46–55	198	33.5
56–65	113	19.1
65+	18	3
No reply	1	0.2

As shown in Table 2, 38.2% of the respondents spent more than 20 years dedicated to research.

Table 2. Years involved in research

	Number	Percentage
<5 years	83	14
5–10 years	81	13.7
11–15 years	85	14.4
16–20 years	110	18.6
>20 years	226	38.2
Never involved	6	1

The most frequent work disciplines were the social sciences (35.3%) and natural sciences (24.5%) (Table 3).

The low response rate in the engineering disciplines is due to the fact that in Spain these studies are taught within the polytechnic universities. The low participation in Health Sciences is due to the fact that only teachers of non-clinical subjects have been included in the sample. The percentage of participation in the Humanities is surprising, suggesting that these researchers are using more data-based methodologies.

Table 3. Work discipline

	Number	Percentage
Agricultural science	13	2.2
Engineering	42	7.1
Humanities	71	13.1
Medical and health	66	11.1
Natural science	145	24.5
Social science	209	35.3
Other	45	7.6

3.2 Type of Data Files and Storing Devices

As shown in Fig. 1, the five most widely used file types were:

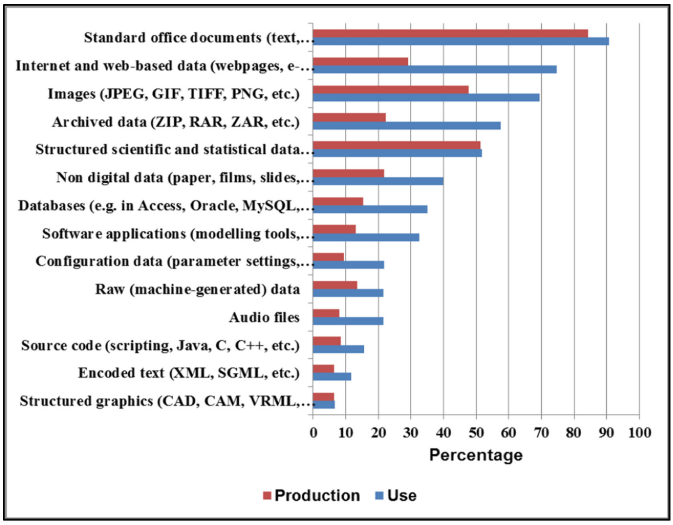


Fig. 1. Type of data files used and produced in research

- Standard office documents (text, spreadsheets, presentations) with 90.7% of respondents.
- Internet and web-based data (webpages, e-mails, blogs, social network data) with 74.8% of respondents.
- Images (JPEG, GIF, TIFF, PNG) with 69.5% of respondents.
- Archived data (ZIP, RAR, ZAR) with 57.5% of respondents.
- Structured scientific and statistical data (such as SPSS, GIS) with 51.9% of respondents.

When asked the type of data they produce in their research, most respondents indicate: standard office documents (84.3%), structured scientific and statistical data (51.3%) and images (47.7%).

With respect to the volume of data they use, the majority of respondents usually used data volumes on the order of GB (44.8%) and MB (45.2%). Only 47 respondents (8%) used data volumes of the order of terabytes. Volumes of data were very similar to those they produced in their research: GB (37.9%) and MB (53.3%).

For most of the respondents (94.3%), the most common way of storing the data generated in the research was to use their own devices (computer, tablet, external drive) or in the cloud (50.3%). Only 17.8% used central servers/repositories of the university or other outside repositories (10%).

3.3 Data Management Plan (DMP)

Figure 2 presents the opinions of academic staff and research students on various aspects related to DMP. A strong majority, 80.4% (n = 591), of the respondents were not aware if their institutions had a DMP and only 10% knew that their institutions had a DMP. Seventy one point nine percent of respondents had never used a DMP in their research, compared to 10% who had used it. This result was consistent with the 8% of respondents who had used a DMP for their current research and 72.6% who said there are not using it. When asked if they thought a DMP actually helps researchers in managing research data, 31.9% of the respondents gave an affirmative answer, compared to 63.8% who did not know.

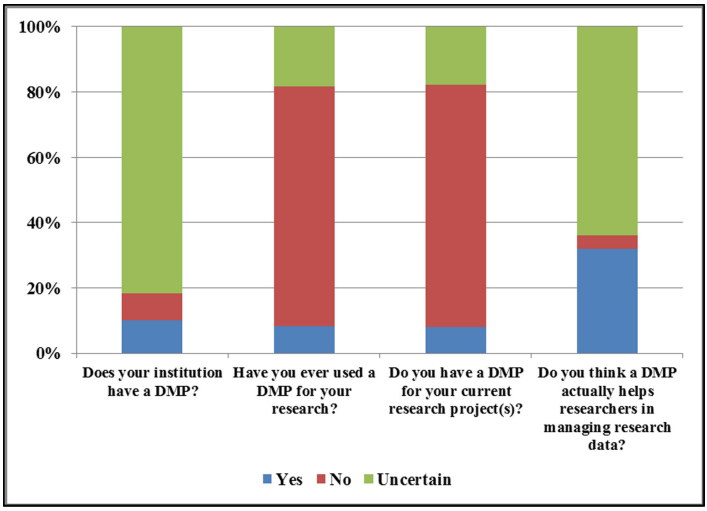


Fig. 2. Opinions about DMP

3.4 Knowledge and Use of Metadata

The survey contains a set of questions about the additional information that researchers include in their research data file from their research.

With respect to additional information, 43.1% of respondents included administrative information (e.g. creator, date of creation, file name, access terms/restrictions,); 41.6% included discovery information (e.g. creator, funding body, project title, project ID, keywords); 23% include technical information (for example, file format, file size, software/hardware needed to use the data,); 24.5% include description of the data file (e.g. file/data structure, field tags/descriptions, application rules,) and 72.6% do not assign additional information to their research data.

Figure 3 summarizes respondents' opinions on metadata. 55.1% of the respondents were familiar with the term "metadata", compared to 39.9% who did not know. Fifty three point five percent of the respondents thought that formal training on metadata would be useful for managing research data and 37.5% did not know. Only 10.9% of the respondents answered that their universities had a prescribed metadata set for uploading data to a repository while 86.1% did not know. Fourteen percent of respondents believed that their research community used/recommended a standard file naming system; where as 52.1% were not aware of it. Only 4.9% of respondents answered that their universities had a standard/consistent file naming system, while 77.7% did not know.

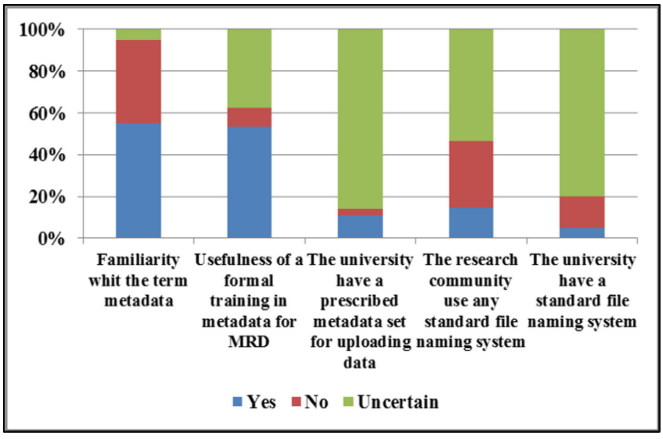


Fig. 3. Opinions about metadata

Figure 4 presents the results on the use of metadata and other ways of identifying data files. Questions have five possible answers: never, rarely, sometimes, often and almost always. Only 10.5% of respondents "often or almost always" used a metadata standard to tag their data. A similar level of response is obtained when asked if they used datasets that are tagged with standard metadata (9.2% indicate often or almost always). Seventeen point eight percent of respondents "often or almost always" used

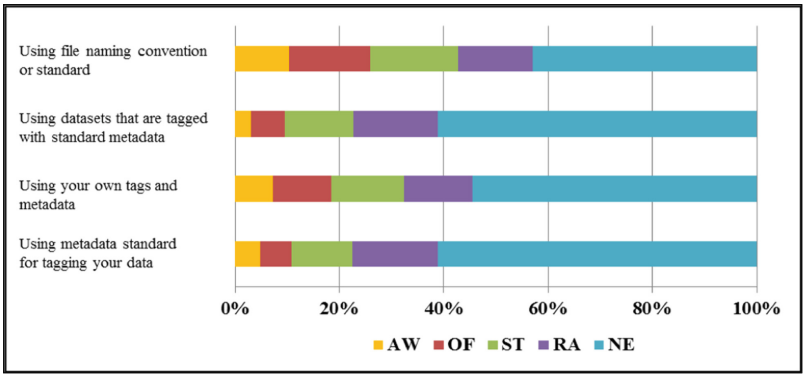


Fig. 4. Use of metadata. AW (Almost Always), OF (Often), ST (Sometimes), RA (Rarely), NE (Never)

their own/in-house (their research team) tags and metadata. Regarding the use of a file naming convention or standard, 25.1% of respondents used it "often or almost always".

3.5 Data Literacy Training

In this section of the questionnaire respondents were asked about their training in some aspects of data management and how they would like to be trained in this matter. Figure 5 presents formal training of the respondents on several issues of data management. Respondents (20.1%) indicated that they only received formal training in data citation styles. Training on other issues, such as Data Management Plan, only wAS provided to 4.6%. Similar results were seen with regards to training on metadata (5.1%), consistent file naming (1.7%), and version control of data sets (2%).

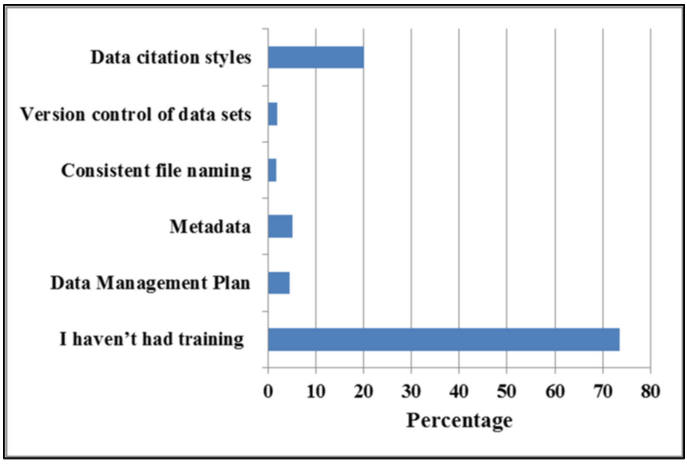


Fig. 5. Formal training received in several issues of data management

Sixty one point four percent of the respondents said they would like to received training on a Data Management Plan, 49.2% on metadata, 42.3% on consistent file naming, 38.9% on version control of data sets, and 37.7% on data citation styles. Only 23.7% of respondents were not interested in receiving training in the above issues.

4 Discussion and Conclusions

This study presents some limitations, among which was the low representation of respondents from University Burgos and the University Rey Juan Carlos. However, the consistency in the results obtained with the different questions may allow us to elaborate on proposals to improve Research Data Management levels at these universities. The results point out the existence of some weak points, such as the current lack of use of DMP in their research activities, such use will need to change in the future when the application of a DMP will become a necessary requirement. Another example was related to the knowledge and use of metadata, where researchers knew the term moderately well, but more than half thought that more training was needed. Similarly this points to the need of a greater involvement of universities in standardization and training activities.

The survey detects the need for strong emphasis on continuing training: more than 60% of respondents agreed to receive training in DMP and more than 40% on metadata and consistent file naming.

In conclusion, according to the results of this survey we have detected a growing need among academic staff and research students for data management literacy skills. In our opinion, the university should offer formal training appropriate to the needs of academic researchers in the different disciplines along with to research students.

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